

A Finite-Horizon Dynamic Reputation Model with Hidden Product Choice

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We study a finite-horizon reputation problem in which a seller privately chooses between a low-cost product and a high-quality product while a user learns only from realized binary outcomes. The user treats the seller as a single Bernoulli arm and applies Thompson sampling to decide whether to engage with the seller or with a known outside option. The seller chooses hidden products to maximize total expected profit over a finite horizon. The resulting dynamic program is indexed by calendar time and by the user's posterior sufficient statistics. The optimal product choice is governed by the marginal continuation value of a success relative to a failure, which gives a simple threshold condition for using the costly product. We also derive a deterministic fluid approximation that replaces random posterior counts by their expected drifts. The fluid formulation preserves the same reputation-investment threshold and provides a transparent long-run benchmark for the quality mix needed to sustain demand.

Key words: dynamic reputation; hidden actions; finite-horizon dynamic programming; fluid approximation; Thompson sampling

1. Model Formulation

Consider a discrete-time horizon $t = 1, \dots, T$. The environment consists of a single user and two sellers, denoted by A and B . At each period t , the user selects a seller, denoted by $y_t \in \{A, B\}$. The user's realized utility is a binary random variable $u_t \in \{0, 1\}$.

Seller B offers a transparent outside option. If the user selects Seller B, the realized utility is Bernoulli with known success probability $p_0 \in (0, 1)$:

$$\mathbb{P}(u_t = 1 \mid y_t = B) = p_0. \quad (1)$$

Seller A has access to two products, indexed by $\mathcal{X} = \{1, 2\}$. When the user selects Seller A, the seller privately chooses a product $x_t \in \mathcal{X}$. Product $i \in \mathcal{X}$ generates a successful outcome with probability p_i and imposes cost c_i on Seller A:

$$\mathbb{P}(u_t = 1 \mid y_t = A, x_t = i) = p_i. \quad (2)$$

We assume

$$0 < p_1 < p_2 \leq 1, \quad c_1 < c_2, \quad (3)$$

so product 2 is more likely to satisfy the user but is also more costly. Whenever Seller A is chosen, it receives a fixed revenue $R > 0$. Seller A's period profit is

$$\pi_t^A = \begin{cases} R - c_{x_t} & \text{if } y_t = A, \\ 0 & \text{if } y_t = B. \end{cases} \quad (4)$$

2. User Learning and Demand

The product chosen by Seller A is hidden from the user. The product affects the distribution of realized utility, but the user observes only the seller selected and the binary realized outcome.

ASSUMPTION 1 (User's Misspecified Belief). *The user observes u_t after choosing Seller A but does not observe the product $x_t \in \mathcal{X}$. The user therefore treats Seller A as a single product with an unknown Bernoulli success probability $\theta \in [0, 1]$.*

The user learns about θ through Bayesian updating. Let S_t and F_t denote the cumulative number of successes and failures generated by Seller A up to the beginning of period t :

$$S_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 1\}, \quad (5)$$

$$F_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 0\}. \quad (6)$$

The user starts from the uniform prior $\theta \sim \text{Beta}(1, 1)$. Given state (S_t, F_t) , the posterior belief is

$$\theta \mid S_t, F_t \sim \text{Beta}(1 + S_t, 1 + F_t). \quad (7)$$

The corresponding posterior mean is

$$m(S_t, F_t) = \frac{S_t + 1}{S_t + F_t + 2}. \quad (8)$$

The user follows a Thompson sampling heuristic. At the beginning of period t , the user samples

$$\tilde{\theta}_t \sim \text{Beta}(1 + S_t, 1 + F_t). \quad (9)$$

Since the outside option has known success probability p_0 , the user chooses Seller A if and only if the sampled index exceeds p_0 :

$$y_t = \begin{cases} A & \text{if } \tilde{\theta}_t \geq p_0, \\ B & \text{if } \tilde{\theta}_t < p_0. \end{cases} \quad (10)$$

Seller A observes the user's choices and realized outcomes, and can therefore track (S_t, F_t) . For a state (S, F) , define the demand probability

$$D(S, F) \triangleq \mathbb{P}(\tilde{\theta}_t \geq p_0 \mid S_t = S, F_t = F) = \int_{p_0}^1 \frac{\theta^S (1 - \theta)^F}{B(1 + S, 1 + F)} d\theta, \quad (11)$$

where $B(\cdot, \cdot)$ is the Beta function. This probability is the chance that Seller A has an opportunity to earn revenue and update the user's belief in the current period.

3. Finite-Horizon Seller Problem

Seller A chooses hidden products to maximize total expected profit over the finite horizon. Let $V_t(S, F)$ denote the optimal expected continuation profit from periods $t, \dots, T$ when the user's belief state at the beginning of period t is (S, F) . The terminal condition is

$$V_{T+1}(S, F) = 0 \quad \text{for all reachable } (S, F). \quad (12)$$

At state (S, F) in period t , the user chooses Seller A with probability $D(S, F)$. If the user chooses Seller B, Seller A earns zero profit and the belief state remains (S, F) . If the user chooses Seller A and the seller uses product $x \in \{1, 2\}$, Seller A earns $R - c_x$ and the next state is $(S + 1, F)$ after a success and $(S, F + 1)$ after a failure. The finite-horizon Bellman equation is

$$\begin{aligned} V_t(S, F) = & [1 - D(S, F)]V_{t+1}(S, F) \\ & + D(S, F) \max_{x \in \{1, 2\}} \{R - c_x + p_x V_{t+1}(S + 1, F) + (1 - p_x)V_{t+1}(S, F + 1)\}. \end{aligned} \quad (13)$$

For each product x , define the period- t action value conditional on the user choosing Seller A:

$$Q_{t,x}(S, F) \triangleq R - c_x + p_x V_{t+1}(S + 1, F) + (1 - p_x)V_{t+1}(S, F + 1). \quad (14)$$

The product choice is made by comparing $Q_{t,1}$ and $Q_{t,2}$. Define the marginal reputation value of a success relative to a failure by

$$M_t(S, F) \triangleq V_{t+1}(S + 1, F) - V_{t+1}(S, F + 1). \quad (15)$$

Then

$$Q_{t,2}(S, F) - Q_{t,1}(S, F) = -(c_2 - c_1) + (p_2 - p_1)M_t(S, F). \quad (16)$$

Therefore, product 2 is optimal at period t and state (S, F) if and only if

$$c_2 - c_1 \leq (p_2 - p_1)M_t(S, F). \quad (17)$$

Equivalently,

$$M_t(S, F) \geq \frac{c_2 - c_1}{p_2 - p_1}. \quad (18)$$

The demand probability $D(S, F)$ scales the expected value of acting in the calendar period, while the product choice itself is governed by the value of improving the user's future belief. A good signal is valuable when it increases future demand enough to justify the incremental cost of product 2.

4. Fluid Approximation

The exact dynamic program retains the full distribution of the posterior count process. We next formulate a deterministic fluid approximation that replaces the random count increments by their conditional expected drifts. This approximation is useful for identifying the economic forces behind the policy and for studying long horizons without enumerating the triangular state space. It does not change the user's information: the user continues to infer Seller A's quality only from observed successes and failures and never observes the seller's hidden product choice.

4.1. Deterministic Count Dynamics

Let $\tau \in [0, T]$ denote continuous calendar time, and let $s(\tau)$ and $f(\tau)$ be the fluid success and failure counts. We extend the demand function in eq. (11) to nonnegative real counts:

$$\mathcal{D}(s, f) \triangleq \int_{p_0}^1 \frac{\theta^s (1 - \theta)^f}{B(1 + s, 1 + f)} d\theta. \quad (19)$$

Let $a(\tau) \in [0, 1]$ denote the relaxed intensity of product 2. It can be interpreted as the fraction of otherwise identical fluid units assigned to product 2, or as time mixing between the two products.

Define

$$q(a) \triangleq p_1 + a(p_2 - p_1), \quad (20)$$

$$c(a) \triangleq c_1 + a(c_2 - c_1). \quad (21)$$

The fluid state then evolves according to

$$\dot{s}(\tau) = \mathcal{D}(s(\tau), f(\tau))q(a(\tau)), \quad (22)$$

$$\dot{f}(\tau) = \mathcal{D}(s(\tau), f(\tau))[1 - q(a(\tau))], \quad (23)$$

with $s(0) = f(0) = 0$. The factor $\mathcal{D}(s, f)$ is the fluid rate at which Seller A is selected and consequently receives a new observation. Conditional on selection, $q(a)$ and $1 - q(a)$ are the expected success and failure rates.

For an admissible measurable control $a(\cdot)$, the fluid profit is

$$J^{\text{fl}}(a) = \int_0^T \mathcal{D}(s(\tau), f(\tau))[R - c(a(\tau))] d\tau. \quad (24)$$

The fluid control problem is to maximize eq. (24) subject to eqs. (22) and (23). This is a deterministic mean-drift closure: in general, $\mathbb{E}[D(S_t, F_t)] \neq D(\mathbb{E}[S_t], \mathbb{E}[F_t])$, so the fluid model is an approximation rather than an exact reformulation of the stochastic problem.

4.2. Fluid Hamilton–Jacobi–Bellman Equation

Let $\mathcal{V}(\tau, s, f)$ denote the optimal fluid profit from time τ to T . Its terminal condition is

$$\mathcal{V}(T, s, f) = 0. \quad (25)$$

The Hamilton–Jacobi–Bellman equation is

$$\begin{aligned} -\partial_\tau \mathcal{V}(\tau, s, f) = \mathcal{D}(s, f) \max_{a \in [0,1]} \{ & R - c(a) + q(a) \partial_s \mathcal{V}(\tau, s, f) \\ & + [1 - q(a)] \partial_f \mathcal{V}(\tau, s, f) \}. \end{aligned} \quad (26)$$

Write $\Delta p = p_2 - p_1 > 0$ and $\Delta c = c_2 - c_1 > 0$. The terms in the Hamiltonian that depend on a are

$$a [-\Delta c + \Delta p (\partial_s \mathcal{V}(\tau, s, f) - \partial_f \mathcal{V}(\tau, s, f))]. \quad (27)$$

Consequently, although the fluid control allows mixing, an optimal control is bang-bang except at points of indifference:

$$a^*(\tau, s, f) = \begin{cases} 1, & \Delta p (\partial_s \mathcal{V} - \partial_f \mathcal{V}) > \Delta c, \\ 0, & \Delta p (\partial_s \mathcal{V} - \partial_f \mathcal{V}) < \Delta c, \\ \text{any value in } [0, 1], & \Delta p (\partial_s \mathcal{V} - \partial_f \mathcal{V}) = \Delta c. \end{cases} \quad (28)$$

Thus the fluid model preserves the threshold structure of eq. (17): product 2 is used when the fluid shadow value of replacing a failure by a success is at least $\Delta c / \Delta p$.

4.3. Effective Observations and Posterior Mean

The fluid dynamics become especially transparent after transforming the state. Let

$$n(\tau) = s(\tau) + f(\tau), \quad m(\tau) = \frac{s(\tau) + 1}{n(\tau) + 2}. \quad (29)$$

Here n is the effective number of observations from Seller A and m is the posterior mean implied by the fluid counts. Define

$$\widehat{\mathcal{D}}(n, m) = \mathbb{P}(\text{Beta}((n+2)m, (n+2)(1-m)) \geq p_0). \quad (30)$$

Using eqs. (22) and (23), the transformed dynamics are

$$\dot{n}(\tau) = \widehat{\mathcal{D}}(n(\tau), m(\tau)), \quad (31)$$

$$\dot{m}(\tau) = \frac{\widehat{\mathcal{D}}(n(\tau), m(\tau))}{n(\tau) + 2} [q(a(\tau)) - m(\tau)]. \quad (32)$$

Equation (32) isolates two forces. The posterior mean moves toward the quality $q(a)$ delivered by the seller, while its speed of adjustment declines as $1/(n+2)$. Demand controls the speed of calendar time at which learning occurs but not the direction of belief updating.

If $\mathcal{W}(\tau, n, m)$ denotes the value function in the transformed state, then

$$\partial_s \mathcal{V} - \partial_f \mathcal{V} = \frac{1}{n+2} \partial_m \mathcal{W}. \quad (33)$$

The fluid product-2 rule can therefore be written as

$$\frac{\Delta p}{n+2} \partial_m \mathcal{W}(\tau, n, m) \geq \Delta c. \quad (34)$$

This representation makes belief rigidity explicit. As the number of observations grows, one additional success has a smaller effect on the posterior mean, so the shadow value $\partial_m \mathcal{W}$ must be correspondingly larger to justify product 2.

4.4. Long-Run Quality-Mix Benchmark

Suppose the relaxed control is held constant at a and demand remains positive. Dividing eq. (32) by eq. (31) yields

$$\frac{dm}{dn} = \frac{q(a) - m}{n+2}. \quad (35)$$

Starting from the uniform prior, $n(0) = 0$ and $m(0) = 1/2$, the solution is

$$m(n) = q(a) + \frac{1 - 2q(a)}{n+2}. \quad (36)$$

Hence the posterior mean converges to the average quality $q(a)$. When $p_1 < p_0 \leq p_2$, the boundary product-2 intensity that keeps this limiting mean at the outside-option quality is

$$a_{\text{bdry}}(p_0) = \frac{p_0 - p_1}{p_2 - p_1}. \quad (37)$$

If $p_0 \leq p_1$, product 1 alone has quality at least as high as the outside option, so the analogous boundary intensity is zero. If $p_0 > p_2$, no feasible quality mix can prevent the posterior mean from eventually falling below p_0 .

The boundary in eq. (37) is a sustainability benchmark, not by itself an optimal stationary policy. Posterior concentration implies

$$\hat{\mathcal{D}}(n, m(n)) \longrightarrow \begin{cases} 1, & q(a) > p_0, \\ 1/2, & q(a) = p_0, \\ 0, & q(a) < p_0. \end{cases} \quad (38)$$

Thus a control strictly above $a_{\text{bdry}}(p_0)$ is needed for demand to converge to one when $p_1 < p_0 < p_2$, whereas the exact boundary produces a singular regime in which the concentrated posterior remains centered at the user's cutoff. For the computational values $p_1 = 0.35$ and $p_2 = 0.80$, the boundary intensities are $1/3$ at $p_0 = 0.50$ and $7/9$ at $p_0 = 0.70$. Product 1 is already sufficient at $p_0 \in \{0.10, 0.30\}$, while demand cannot be sustained asymptotically at $p_0 = 0.90$ even under product 2.

4.5. Role and Limitations of the Fluid Model

The fluid approximation provides three useful benchmarks. First, it retains the same quality-investment threshold as the exact dynamic program. Second, it separates the rate at which observations arrive from the rate at which each observation changes the posterior mean. Third, it identifies the minimum long-run quality mix needed to keep Seller A competitive with the outside option.

The approximation deliberately suppresses path dispersion. This matters most near $m = p_0$, where Thompson-sampling demand changes rapidly and eq. (38) is discontinuous in limiting quality. A diffusion approximation would be needed to retain the order- $\sqrt{n}$ fluctuations around that boundary. Accordingly, the fluid solution should be evaluated as a long-horizon benchmark and compared against the exact backward-induction policy and stochastic simulation rather than treated as an exact substitute.

5. Exact Backward Induction

At the beginning of period t , the reachable states satisfy

$$\mathcal{S}_t = \{(S, F) \in \mathbb{N}_0^2 : S + F \leq t - 1\}. \quad (39)$$

Thus the finite-horizon problem can be solved exactly by backward induction over the triangular state space. Starting from eq. (12), the algorithm computes V_t from eq. (13) for $t = T, T - 1, \dots, 1$. The implementation stores only the current and next value arrays during the backward pass. It also records early-period policy slices and the marginal reputation object $M_t(S, F)$ for diagnostic analysis.

The policy diagnostics focus on four state-dependent objects:

1. the marginal reputation value $M_t(S_t, F_t)$;
2. the demand probability $D(S_t, F_t)$;
3. the cumulative number of Seller A observations $S_t + F_t$;
4. the posterior mean $(S_t + 1)/(S_t + F_t + 2)$.

These objects separate the user's propensity to choose Seller A, the rigidity of the user's belief, the location of the posterior belief, and the value to Seller A of generating a success rather than a failure.

6. Interpretation

The central object in this finite-horizon problem is $M_t(S, F)$. It measures how much additional continuation value Seller A obtains from a success rather than a failure at the next belief update. Product 2 is used exactly when this marginal reputation value is large enough to pay for its incremental cost. The demand probability $D(S, F)$ determines how often the seller receives the

opportunity to earn revenue and update the state, while $S + F$ and the posterior mean determine how responsive the user's belief is to future outcomes. This decomposition makes the finite-horizon policy transparent: the costly product is not chosen because it raises current demand, but because it can improve the future reputation state when the reputational return is high enough. The fluid model preserves this logic through the shadow-value condition eq. (34) and complements it with a tractable benchmark for the long-run quality mix.